Structured notes: Module 3, electromagnetic waves

Griffiths, *Introduction to Electrodynamics* (5th ed.), §8.1.2 and Chapter 9. Figures are redrawn for these notes, not copied from the book; the book figure number is in each caption. Sections 4 (§9.3.3) and 5 (§9.4, §9.5) are background, beyond the syllabus list. Problems are not included.

Section 1: energy and Poynting's theorem (Griffiths 8.1.2)

8.1.2 Poynting's theorem

Source: book-sources/m3-s812-poynting.html, Griffiths pp. 363–367. Equations 8.5–8.12 and Example 8.1 in full. Problems 8.1–8.4 are not included.

Prerequisites

- Divergence theorem; product rule for $\nabla \cdot (\mathbf{E} \times \mathbf{B})$ (pre-math M13, M14).
- Faraday's law $\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$ and the Ampère–Maxwell law (Module 2, Section 7.3).
- Field energy densities from Eqs. 2.45 and 7.44.

1. Energy density of the fields

The work to assemble a static charge distribution is $W_e = \frac{\epsilon_0}{2} \int E^2 \, d\tau$ (Eq. 2.45). The work to establish currents against the back emf is $W_m = \frac{1}{2\mu_0} \int B^2 \, d\tau$ (Eq. 7.44). Taking the integrands as densities gives the energy per unit volume:

RESULT 8.1.1: ENERGY DENSITY

$$u = \frac{1}{2}\left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2\right) \tag{8.5}$$

2. Rate of work on the charges

The Lorentz force on a charge q does work $dW = \mathbf{F} \cdot d\mathbf{l} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} \, dt = q\mathbf{E} \cdot \mathbf{v} \, dt$, since $\mathbf{v} \times \mathbf{B} \perp \mathbf{v}$. With $q \to \rho \, d\tau$ and $\rho \mathbf{v} \to \mathbf{J}$, the rate of work on all charges in volume $\mathcal{V}$ is

$$\frac{dW}{dt} = \int_{\mathcal{V}} (\mathbf{E} \cdot \mathbf{J}) \, d\tau \tag{8.6}$$

Hence $\mathbf{E} \cdot \mathbf{J}$ is the power per unit volume delivered to the charges.

3. Elimination of $\mathbf{J}$

By the Ampère–Maxwell law, $\mathbf{J}$ is replaced by fields:

$$\mathbf{E} \cdot \mathbf{J} = \frac{1}{\mu_0} \mathbf{E} \cdot (\nabla \times \mathbf{B}) - \epsilon_0 \, \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t}$$

The product rule $\nabla \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{B})$, with Faraday's law, gives

$$\mathbf{E} \cdot (\nabla \times \mathbf{B}) = -\mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} - \nabla \cdot (\mathbf{E} \times \mathbf{B})$$

Each time derivative of a square is written as

$$\begin{aligned} \mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} &= \frac{1}{2} \frac{\partial}{\partial t}(B^2) \\ \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} &= \frac{1}{2} \frac{\partial}{\partial t}(E^2) \end{aligned} \tag{8.7}$$

Combining,

$$\mathbf{E} \cdot \mathbf{J} = -\frac{1}{2} \frac{\partial}{\partial t}\left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2\right) - \frac{1}{\mu_0} \nabla \cdot (\mathbf{E} \times \mathbf{B}) \tag{8.8}$$

4. Poynting's theorem

Inserting Eq. 8.8 into Eq. 8.6 and applying the divergence theorem to the second term:

$$\frac{dW}{dt} = -\frac{d}{dt} \int_{\mathcal{V}} u \, d\tau - \frac{1}{\mu_0} \oint_S (\mathbf{E} \times \mathbf{B}) \cdot d\mathbf{a} \tag{8.9}$$

The work done on the charges equals the decrease of field energy in $\mathcal{V}$, less the energy flowing out through S. This is Poynting's theorem, the work–energy theorem of electrodynamics.

5. Poynting vector

RESULT 8.1.2: ENERGY FLUX DENSITY

$$\mathbf{S} \equiv \frac{1}{\mu_0} (\mathbf{E} \times \mathbf{B}) \tag{8.10}$$

$\mathbf{S}\cdot d\mathbf{a}$ is the energy per unit time crossing $d\mathbf{a}$. The compact form is

$$\frac{dW}{dt} = -\frac{d}{dt}\int_{\mathcal{V}} u\,d\tau - \oint_S \mathbf{S}\cdot d\mathbf{a} \tag{8.11}$$

Footnote 4 (book): Eq. 8.9 applies to closed surfaces only. The power through an open surface is the natural interpretation of $\int \mathbf{S}\cdot d\mathbf{a}$, but not a consequence of Eq. 8.9. The location of energy is not determined in electrodynamics (Section 2.4.4).

6. Local conservation in empty space

With no charge in $\mathcal{V}$, $dW/dt = 0$. From Eq. 8.11, $\int \partial_t u\,d\tau = -\oint \mathbf{S}\cdot d\mathbf{a} = -\int \nabla\cdot\mathbf{S}\,d\tau$. Since $\mathcal{V}$ is arbitrary,

RESULT 8.1.3: CONTINUITY EQUATION FOR ENERGY

$$\frac{\partial u}{\partial t} = -\nabla\cdot\mathbf{S} \tag{8.12}$$

u corresponds to ρ and $\mathbf{S}$ to $\mathbf{J}$ in the charge continuity equation Eq. 8.4.

Electromagnetic energy alone is not conserved in general, because the fields exchange energy with the charges.

7. Example 8.1: resistive wire (book)

Uniform field along a wire of length L and radius a: $E = V/L$. Circumferential field at the surface: $B = \mu_0 I/2\pi a$. Magnitude of the Poynting vector:

$$S = \frac{1}{\mu_0}\cdot\frac{V}{L}\cdot\frac{\mu_0 I}{2\pi a} = \frac{VI}{2\pi a L}$$

$\mathbf{E}\times\mathbf{B}$ is radial, and the direction is inward. Over the lateral surface, of area $2\pi aL$:

$$\int \mathbf{S}\cdot d\mathbf{a} = S(2\pi aL) = VI$$

This agrees with the Joule heating obtained in Section 7.1.1.

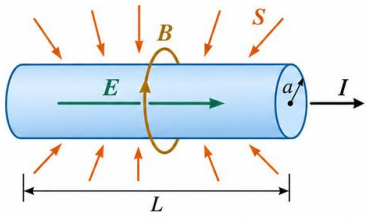


Fig. 8.1 (book): a resistive wire carrying current I; E along the wire, B circumferential, S radially inward.

Problems 8.1–8.4 apply the same theorem to cables, a capacitor gap and a charging sphere. They are outside these notes.

Section 2: waves in one dimension (Griffiths 9.1)

9.1.1–9.1.2 The wave equation and sinusoidal waves

Source: book-sources/m3-sec2-wave1d.html, Griffiths pp. 390–396. Equations 9.1–9.20 in full. Problems 9.1–9.4 are listed, not worked.

Prerequisites

- Chain rule for functions of $z - vt$ (pre-math M7).
- Euler's formula (pre-math M5).
- Newton's second law and small-angle approximations for a string (Plus Two mechanics).

1. Definition and the general form

A wave is a disturbance of a continuous medium that propagates with a fixed shape at constant velocity. Refinements (absorption, dispersion, spreading in two or three dimensions, standing waves) are set aside for the one-dimensional case shown in Fig. 9.1.

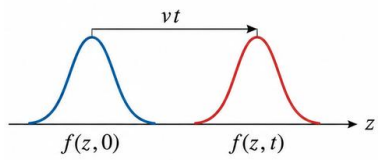


Fig. 9.1 (book): a wave of fixed shape at t = 0 and at a later time, shifted to the right by vt.

With $g(z) \equiv f(z, 0)$, the displacement at z at time t equals the initial displacement at $z - vt$:

$$f(z, t) = f(z - vt, 0) = g(z - vt) \tag{9.1}$$

A function whose dependence on z and t is confined to the combination $z - vt$ is a wave of fixed shape moving in the $+z$ direction at speed v. The functions $f_1 = Ae^{-b(z-vt)^2}$, $f_2 = A\sin[b(z - vt)]$ and $f_3 = A/[b(z - vt)^2 + 1]$ are waves. The functions $f_4 = Ae^{-b(bz^2+vt)}$ and $f_5 = A\sin(bz)\cos(bvt)$ are not.

2. Derivation of the wave equation

Net transverse force on a segment of string of length Δz under tension T, with small angles:

$$\Delta F = T\sin\theta' - T\sin\theta \approx T\frac{\partial^2 f}{\partial z^2}\Delta z$$

Newton's second law with mass per unit length μ gives $\Delta F = \mu\,\Delta z\,\partial^2 f/\partial t^2$. Therefore small transverse disturbances obey

RESULT 9.1: WAVE EQUATION ON A STRING

$$\frac{\partial^2 f}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2} \tag{9.2}$$

$$v = \sqrt{\frac{T}{\mu}} \tag{9.3}$$

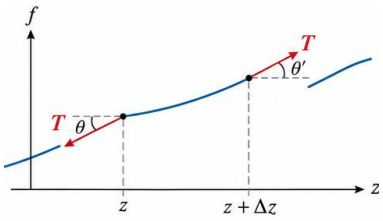


Fig. 9.2 (book): a string segment between z and z + Δz, with tension T at angles θ and θ′.

3. Solutions

Set $u = z - vt$. Then $\partial f/\partial z = dg/du$, $\partial f/\partial t = -v\,dg/du$, $\partial^2 f/\partial z^2 = d^2g/du^2$ and $\partial^2 f/\partial t^2 = v^2\,d^2g/du^2$. Substitution into Eq. 9.2 confirms that any function of the form

$$f(z, t) = g(z - vt) \tag{9.4}$$

is a solution, for any differentiable g. Changing the sign of v gives a wave to the left:

$$f(z, t) = h(z + vt) \tag{9.5}$$

Since Eq. 9.2 is linear, the general solution is the sum:

$$f(z, t) = g(z - vt) + h(z + vt) \tag{9.6}$$

Every solution of Eq. 9.2 takes this form (book). Note 1 of the book observes that the solution class is far larger than for an ordinary second-order ODE, which admits two independent solutions; a PDE has no such restriction.

4. Sinusoidal waves

The sinusoidal wave is

$$f(z,t) = A\cos[k(z - vt) + \delta] \tag{9.7}$$

with A the amplitude, δ the phase constant, and δ/k the delay of the central maximum. Wavelength and period:

$$\lambda = \frac{2\pi}{k} \tag{9.8}$$

$$T = \frac{2\pi}{kv} \tag{9.9}$$

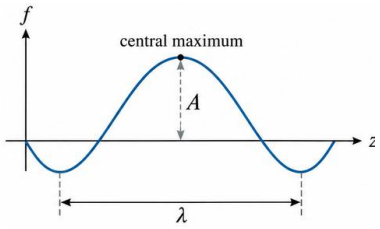


Fig. 9.3 (book): a sinusoidal wave at t = 0, with amplitude A, wavelength λ and the central maximum at δ/k.

> **PLUS TWO RECALL: SPEED OF A WAVE**
>
> **Recall.** From Class 12: speed = frequency × wavelength.
>
> **Re-derive.**
>
> 1. In one period the wave advances by one wavelength: $v = \lambda/T$.
> 2. Frequency is the reciprocal of period: $\nu = 1/T$.
> 3. Hence $\nu = v/\lambda$; using Eq. 9.9 for T gives $\nu = kv/2\pi$.
>
> **Saved.** $\nu = kv/2\pi = v/\lambda = 1/T$ Eq. 9.10.

$$\nu = \frac{kv}{2\pi} = \frac{v}{\lambda} = \frac{1}{T} \tag{9.10}$$

> **RESULT 9.2: ANGULAR FREQUENCY**
>
> $$\omega = 2\pi\nu = kv \tag{9.11}$$
>
> $$f(z,t) = A\cos(kz - \omega t + \delta) \tag{9.12}$$

Leftward propagation: $f = A\cos(kz + \omega t - \delta)$ Eq. 9.13 $= A\cos(-kz - \omega t + \delta)$ Eq. 9.14. Changing the sign of k reverses the direction and leaves the other parameters unchanged.

5. Complex notation

By Euler's formula Eq. 9.15,

$$f(z,t) = \mathrm{Re}\left[Ae^{i(kz - \omega t + \delta)}\right] \tag{9.16}$$

Define $\tilde{f}(z,t) = \tilde{A}e^{i(kz-\omega t)}$ Eq. 9.17 with $\tilde{A} = Ae^{i\delta}$. Then $f = \mathrm{Re}\,\tilde{f}$ Eq. 9.18.

Example 9.1. Waves of equal k and ω add by adding complex amplitudes:

$$A_3 e^{i\delta_3} = A_1 e^{i\delta_1} + A_2 e^{i\delta_2} \tag{9.19}$$

Restriction: the method is linear and does not extend to products, since $\mathrm{Re}(\tilde{f}_1\tilde{f}_2) \neq \mathrm{Re}\tilde{f}_1\,\mathrm{Re}\tilde{f}_2$. Quadratic quantities must use real fields.

6. General waves

Any wave is a superposition of sinusoids over all wave numbers, with k allowed to be negative to include both directions:

$$\tilde{f}(z,t) = \int_{-\infty}^{\infty} \tilde{A}(k)e^{i(kz-\omega t)}\,dk \tag{9.20}$$

The coefficients are given by Fourier theory (Prob. 9.34). The book then restricts all further work to sinusoidal waves, since the behaviour of a general wave follows from it. Problems 9.3 and 9.4 are outside these notes.

9.1.3 Reflection and transmission on a string

*Source: book-sources/m3-sec2-bc.html, Griffiths pp. 396–399. Equations 9.21–9.33 in full. **Background**: beyond the syllabus list (items 9–14 use §9.1.1, §9.1.2 and §9.1.4). Problems 9.5–9.7 not worked.*

Prerequisites

- Sinusoidal waves and complex amplitudes (§9.1.2).
- Continuity of a function and of its first derivative at a point.

1. Setup

Two strings of equal tension T and mass densities μ_1, μ_2 are joined at $z = 0$. Wave speeds: $v_1 = \sqrt{T/\mu_1}$, $v_2 = \sqrt{T/\mu_2}$.

2. Incident, reflected and transmitted waves

$$\tilde{f}_I = \tilde{A}_I e^{i(k_1 z - \omega t)} \quad (z < 0) \tag{9.21}$$

$$\tilde{f}_R = \tilde{A}_R e^{i(-k_1 z - \omega t)} \quad (z < 0) \tag{9.22}$$

$$\tilde{f}_T = \tilde{A}_T e^{i(k_2 z - \omega t)} \quad (z > 0) \tag{9.23}$$

Common frequency ω; wavelength ratio $\lambda_1/\lambda_2 = k_2/k_1 = v_1/v_2$ Eq. 9.24.

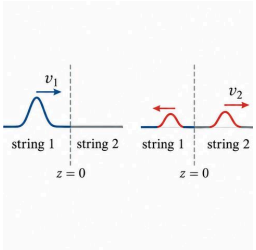


Fig. 9.5 (book): a pulse incident on a knot at z = 0, with the reflected and transmitted pulses.

A finite pulse is a superposition of sinusoids (Eq. 9.20); a single frequency requires infinite extent. The net disturbance is

$$\tilde{f}(z,t) = \begin{cases} \tilde{A}_I e^{i(k_1 z - \omega t)} + \tilde{A}_R e^{i(-k_1 z - \omega t)}, & z < 0 \\ \tilde{A}_T e^{i(k_2 z - \omega t)}, & z > 0 \end{cases} \tag{9.25}$$

3. Matching conditions

RESULT 9.3: JUNCTION CONDITIONS

$$f(0^-, t) = f(0^+, t) \tag{9.26}$$

$$\left.\frac{\partial f}{\partial z}\right|_{0^-} = \left.\frac{\partial f}{\partial z}\right|_{0^+} \tag{9.27}$$

Eq. 9.27 holds for a massless knot; a mismatch would give an infinite acceleration (Fig. 9.6). The same conditions hold for $\tilde{f}$ Eq. 9.28:

$$\begin{aligned} \tilde{f}(0^-, t) &= \tilde{f}(0^+, t) \\ \left.\frac{\partial \tilde{f}}{\partial z}\right|_{0^-} &= \left.\frac{\partial \tilde{f}}{\partial z}\right|_{0^+} \end{aligned} \tag{9.28}$$

4. Amplitudes

Continuity: $\tilde{A}_I + \tilde{A}_R = \tilde{A}_T$. Slope: $k_1(\tilde{A}_I - \tilde{A}_R) = k_2 \tilde{A}_T$. Hence

$$\begin{aligned} \tilde{A}_R &= \frac{k_1 - k_2}{k_1 + k_2} \tilde{A}_I \\ \tilde{A}_T &= \frac{2k_1}{k_1 + k_2} \tilde{A}_I \end{aligned} \tag{9.29}$$

In terms of speeds (via 9.24):

$$\begin{aligned} \tilde{A}_R &= \frac{v_2 - v_1}{v_2 + v_1} \tilde{A}_I \\ \tilde{A}_T &= \frac{2v_2}{v_2 + v_1} \tilde{A}_I \end{aligned} \tag{9.30}$$

With real amplitudes and phases:

$$\begin{aligned} A_R e^{i\delta_R} &= \frac{v_2 - v_1}{v_2 + v_1} A_I e^{i\delta_I} \\ A_T e^{i\delta_T} &= \frac{2v_2}{v_2 + v_1} A_I e^{i\delta_I} \end{aligned} \tag{9.31}$$

Case $v_2 > v_1$*:* no phase change; $A_R = \frac{v_2 - v_1}{v_2 + v_1} A_I$, $A_T = \frac{2v_2}{v_2 + v_1} A_I$ Eq. 9.32.

Case $v_2 < v_1$*:* reflected phase shifted by π ("upside down"); $A_R = \frac{v_1 - v_2}{v_2 + v_1} A_I$, $A_T = \frac{2v_2}{v_2 + v_1} A_I$ Eq. 9.33.

Fixed end ($v_2 \to 0$ relative): $A_R = A_I$, $A_T = 0$, full reflection.

9.1.4 Polarization of transverse waves

Source: book-sources/m3-sec2-polar.html, Griffiths pp. 399–401. Equations 9.34–9.39 in full. Problem 9.8 (circular polarization) described, not worked.

Prerequisites

- Dot product of vectors (pre-math M9, M10).
- Sinusoidal waves (§9.1.2).

1. Transverse and longitudinal waves

In a transverse wave the displacement is perpendicular to the direction of propagation. In a longitudinal (compression) wave the displacement is along it (Fig. 9.7). Sound is longitudinal; electromagnetic waves are transverse.

2. Polarization states

For propagation along z, there are two independent transverse displacement directions. Vertical:

$$\tilde{\mathbf{f}}^{v} = \tilde{A} e^{i(kz-\omega t)} \hat{x} \tag{9.34}$$

Horizontal:

$$\tilde{\mathbf{f}}^{h} = \tilde{A} e^{i(kz-\omega t)} \hat{y} \tag{9.35}$$

General direction in the xy plane, described by a unit vector $\hat{n}$:

$$\tilde{\mathbf{f}} = \tilde{A} e^{i(kz-\omega t)} \hat{n} \tag{9.36}$$

RESULT 9.4: TRANSVERSALITY AND POLARIZATION ANGLE

$$\hat{n} \cdot \hat{z} = 0 \tag{9.37}$$

$$\hat{n} = \cos\theta\, \hat{x} + \sin\theta\, \hat{y} \tag{9.38}$$

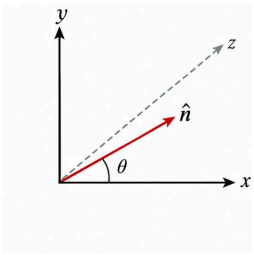


Fig. 9.8c (book): a wave polarized along n̂, at angle θ to the x axis in the xy plane.

Superposition of two orthogonal components:

$$\begin{aligned}\tilde{\mathbf{f}} = {} & (\tilde{A}\cos\theta) e^{i(kz-\omega t)} \hat{x} \\ & + (\tilde{A}\sin\theta) e^{i(kz-\omega t)} \hat{y}\end{aligned} \tag{9.39}$$

3. Remarks

- Reversing $\hat{n}$ and advancing the phase constant by $180°$ gives the same wave (footnote 4).
- Equal amplitudes with a $90°$ phase difference give circular polarization. Problem 9.8 covers the sense of rotation and how to produce each sense; the optics names are right and left circular polarization.
- The book does not use a complex $\hat{n}$ for elliptical polarization.

Section 3: electromagnetic waves in vacuum (Griffiths 9.2)

9.2.1 The wave equation for E and B

Source: book-sources/m3-sec3-wave-eq.html and the end of m3-sec2-polar.html, Griffiths pp. 401–402. Equations 9.40–9.43 in full.

Prerequisites

- Maxwell’s equations in vacuum (Module 2, Section 7.3).
- Identity $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$ (pre-math M13).

1. Maxwell’s equations in empty space

$$\text{(i)}\ \nabla \cdot \mathbf{E} = 0$$

$$\text{(iii)}\ \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\text{(ii)}\ \nabla\cdot\mathbf{B}=0 \qquad \text{(iv)}\ \nabla\times\mathbf{B}=\mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t} \tag{9.40}$$

2. Uncoupling

The curl of (iii), using (iv) and (i):

$$\begin{aligned}\nabla\times(\nabla\times\mathbf{E}) &= \nabla(\nabla\cdot\mathbf{E})-\nabla^2\mathbf{E}\\ &= -\mu_0\epsilon_0\frac{\partial^2\mathbf{E}}{\partial t^2}\end{aligned}$$

The curl of (iv), using (iii) and (ii), gives the same form for $\mathbf{B}$. Hence

RESULT 9.6: DECOUPLED WAVE EQUATIONS

$$\nabla^2\mathbf{E}=\mu_0\epsilon_0\frac{\partial^2\mathbf{E}}{\partial t^2} \qquad \nabla^2\mathbf{B}=\mu_0\epsilon_0\frac{\partial^2\mathbf{B}}{\partial t^2} \tag{9.41}$$

These are second-order equations. Each component satisfies $\nabla^2 f=(1/v^2)\,\partial^2 f/\partial t^2$.

3. Speed

PLUS TWO RECALL: SPEED OF LIGHT FROM ϵ_0 AND μ_0

Recall. From Class 12: electromagnetic waves travel in vacuum at $c=1/\sqrt{\epsilon_0\mu_0}$.

Re-derive.

1. For a wave $f(z-vt)$, Eq. 9.2 gives $\partial^2 f/\partial z^2=(1/v^2)\partial^2 f/\partial t^2$.
2. Comparing with Eq. 9.41 gives $1/v^2=\mu_0\epsilon_0$.
3. So $v=1/\sqrt{\mu_0\epsilon_0}$.

Saved. $c=1/\sqrt{\epsilon_0\mu_0}$ Eq. 9.43.

RESULT 9.7: SPEED IN VACUUM

$$v=\frac{1}{\sqrt{\epsilon_0\mu_0}}=3.00\times10^8\ \text{m/s} \tag{9.42}$$

$$c=\frac{1}{\sqrt{\epsilon_0\mu_0}} \tag{9.43}$$

The number is the measured speed of light, so light is an electromagnetic wave (footnote 6 of the book).

4. Role of the displacement current

The term $\mu_0\epsilon_0\,\partial\mathbf{E}/\partial t$ in (iv) is what makes waves possible. Without it there is no wave equation. The constants ϵ_0 and μ_0 are measured with static charges and steady currents, which have nothing to do with light, yet they fix c.

9.2.2 Monochromatic plane waves

Source: book-sources/m3-sec3-plane.html, Griffiths pp. 402–406. Equations 9.44–9.53, Table 9.1 and Example 9.2 in full. Problem 9.9 not worked. Figures redrawn.

Prerequisites

- Complex exponentials, real part (§9.1.2).
- Faraday's law (Eq. 9.40, iii); dot and cross products (pre-math M8–M10).

1. Definitions

Monochromatic: a single frequency ω. Plane wave: uniform over each plane perpendicular to the direction of travel z (Fig. 9.9). Then

$$\tilde{\mathbf{E}}(z,t)=\tilde{\mathbf{E}}_0 e^{i(kz-\omega t)} \qquad \tilde{\mathbf{B}}(z,t)=\tilde{\mathbf{B}}_0 e^{i(kz-\omega t)} \tag{9.44}$$

with $\omega=ck$.

Table 9.1 (book), the electromagnetic spectrum, is reproduced in the study version.

2. Transversality and amplitude relations

Substituting Eq. 9.44 into $\nabla \cdot \mathbf{E} = 0$ and $\nabla \cdot \mathbf{B} = 0$:

$$(\tilde{E}_0)_z = (\tilde{B}_0)_z = 0 \tag{9.45}$$

Faraday's law then gives

$$\begin{aligned} -k(\tilde{E}_0)_y &= \omega(\tilde{B}_0)_x \\ k(\tilde{E}_0)_x &= \omega(\tilde{B}_0)_y \end{aligned} \tag{9.46}$$

$$\tilde{\mathbf{B}}_0 = \frac{k}{\omega}(\hat{z} \times \tilde{\mathbf{E}}_0) \tag{9.47}$$

PLUS TWO RECALL: TRANSVERSE, IN-PHASE FIELDS

Recall. From Class 12: $\mathbf{E}$ and $\mathbf{B}$ are in phase, mutually perpendicular, and perpendicular to the direction of travel.

Re-derive.

1. Take $\tilde{\mathbf{E}}_0 = E_0\hat{x}$. By Eq. 9.47, $\tilde{\mathbf{B}}_0 = (k/\omega)E_0(\hat{z} \times \hat{x}) = (k/\omega)E_0\hat{y}$.
2. Faraday's law gives $kE_0 = \omega B_0$.
3. Both fields carry the phase $kz - \omega t$, so they are in phase.

Saved. $B_0 = \frac{k}{\omega}E_0 = \frac{E_0}{c}$ Eq. 9.48.

RESULT 9.8: AMPLITUDE RELATION

$$B_0 = \frac{k}{\omega}E_0 = \frac{1}{c}E_0 \tag{9.48}$$

3. Example 9.2

For $\mathbf{E}$ along x, $\mathbf{B}$ is along y. Real fields:

$$\begin{aligned} \mathbf{E}(z,t) &= E_0\cos(kz - \omega t + \delta)\hat{x}, \\ \mathbf{B}(z,t) &= \frac{1}{c}E_0\cos(kz - \omega t + \delta)\hat{y} \end{aligned} \tag{9.49}$$

Polarization is named by the direction of $\mathbf{E}$ (Fig. 9.10).

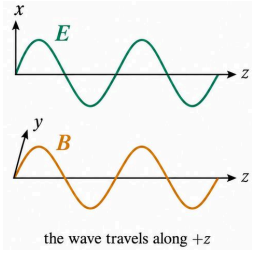


Fig. 9.10 (book): monochromatic plane wave with E along x and B along y, traveling along z.

4. Propagation vector

For direction $\hat{k}$, the propagation vector $\mathbf{k} = k\hat{k}$ and the phase $\mathbf{k} \cdot \mathbf{r}$ (Fig. 9.11):

$$\begin{aligned} \tilde{\mathbf{E}}(\mathbf{r},t) &= \tilde{E}_0 e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}\hat{n} \\ \tilde{\mathbf{B}}(\mathbf{r},t) &= \frac{1}{c}\tilde{E}_0 e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}(\hat{k} \times \hat{n}) = \frac{1}{c}\hat{k} \times \tilde{\mathbf{E}} \end{aligned} \tag{9.50}$$

Transversality: $\hat{n} \cdot \hat{k} = 0$ Eq. 9.51. Real fields:

$$\mathbf{E}(\mathbf{r},t) = E_0\cos(\mathbf{k}\cdot\mathbf{r} - \omega t + \delta)\hat{n} \tag{9.52}$$

$$\mathbf{B}(\mathbf{r},t) = \frac{1}{c}E_0\cos(\mathbf{k}\cdot\mathbf{r} - \omega t + \delta)(\hat{k} \times \hat{n}) \tag{9.53}$$

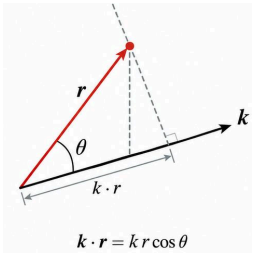


Fig. 9.11 (book): the propagation vector k and the position vector r; the phase is k · r.

Problem 9.9 is outside these notes.

9.2.3 Energy and momentum in electromagnetic waves

Source: book-sources/m3-sec3-energy.html, Griffiths pp. 406–408. Equations 9.54–9.65 in full; footnotes 9–11. Problem 9.10 described; Problems 9.11–9.13 listed, not worked.

Prerequisites

- u and $\mathbf{S}$ from Section 8.1.2 (Eqs. 8.5, 8.10); the momentum density result (Eq. 8.29).
- Average of $\cos^2$ over a period, $\frac{1}{2}$ (pre-math M4).

1. Energy density

$$u = \frac{1}{2}\left(\epsilon_0 E^2 + \frac{1}{\mu_0} B^2\right) \tag{9.54}$$

For a monochromatic plane wave, $B^2 = E^2/c^2 = \mu_0\epsilon_0 E^2$ Eq. 9.55. The two terms are equal:

> **PLUS TWO RECALL: ELECTRIC ENERGY DENSITY**
>
> **Recall.** From Class 12: a field E in vacuum stores $\frac{1}{2}\epsilon_0 E^2$ per unit volume.
>
> **Re-derive.**
>
> 1. The magnetic term of Eq. 9.54 is $\frac{1}{2\mu_0} B^2$.
> 2. With $B = E/c$ and $c^2 = 1/\epsilon_0\mu_0$, this is $\frac{1}{2}\epsilon_0 E^2$.
> 3. The two halves are equal.
>
> **Saved.**

$$u = \epsilon_0 E^2 = \epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta) \tag{9.56}$$

2. Poynting vector of a plane wave

> **RESULT 9.9: ENERGY FLUX**
>
> $$\mathbf{S} = \frac{1}{\mu_0}(\mathbf{E} \times \mathbf{B}) \tag{9.57}$$
>
> $$\mathbf{S} = c\epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta)\hat{z} = cu\,\hat{z} \tag{9.58}$$

The flux equals cu because in time Δt the energy $uAc\Delta t$ crosses an area A.

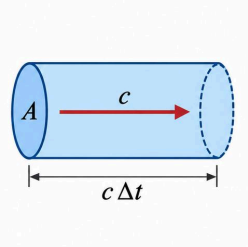


Fig. 9.12 (book): a cylinder of cross section A and length cΔt; the energy in it crosses A in time Δt.

3. Momentum density

$$\mathbf{g} = \frac{1}{c^2}\mathbf{S} \tag{9.59}$$

$$\begin{aligned}\mathbf{g} &= \frac{1}{c}\epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta)\hat{z} \\ &= \frac{1}{c} u\,\hat{z}\end{aligned} \tag{9.60}$$

Footnote 9: $\mathrm{Re}(\tilde{u}) \neq u$ for the complex form, so energy densities are always computed from the real fields.

4. Averages and intensity

Because the period (about 10^{-15} s) is much shorter than any measurement, the averages over a cycle are used (footnote 10: $\langle\cos^2\rangle = \frac{1}{2}$):

$$\langle u \rangle = \frac{1}{2}\epsilon_0 E_0^2 \tag{9.61}$$

$$\langle \mathbf{S} \rangle = \frac{1}{2} c\epsilon_0 E_0^2 \hat{z} \tag{9.62}$$

$$\langle \mathbf{g} \rangle = \frac{1}{2c}\epsilon_0 E_0^2 \hat{z} \tag{9.63}$$

> **RESULT 9.10: INTENSITY**
>
> $$I \equiv \langle S \rangle = \frac{1}{2} c\epsilon_0 E_0^2 \tag{9.64}$$

5. Radiation pressure

$$\begin{aligned}P &= \frac{1}{A}\frac{\Delta p}{\Delta t} \\ &= \frac{1}{2}\epsilon_0 E_0^2 \\ &= \frac{I}{c}\end{aligned} \tag{9.65}$$

With $\Delta p = \langle g \rangle Ac\,\Delta t$ for a perfect absorber. A perfect reflector gives $2I/c$. The qualitative account with the Lorentz force is incomplete (footnote 11; Problem 9.11).

Problem 9.10: sunlight at 1300 W/m^2 on an absorber and on a reflector. Problems 9.11–9.13 are outside these notes.

Section 4: electromagnetic waves in matter (Griffiths 9.3)

9.3.1–9.3.2 Propagation in linear media; normal incidence

Source: book-sources/m3-sec4-linear.html and m3-sec4-oblique.html (Eqs. 9.87–9.89), Griffiths pp. 408–412. Equations 9.66–9.89 in full; footnotes 13–16; Problems 9.14–9.15 described.

Prerequisites

- Vacuum Maxwell equations (Eq. 9.40) and plane-wave relations Eq. 9.46–Eq. 9.48.
- Interface conditions (Eq. 7.74, given as 9.75).
- Permittivity and permeability of matter (pre-physics P12, P13).

1. Maxwell's equations in linear media

$$\begin{aligned} &\text{(i)}\ \nabla\cdot\mathbf{D}=0 \\ &\text{(iii)}\ \nabla\times\mathbf{E}=-\frac{\partial\mathbf{B}}{\partial t} \\ &\text{(ii)}\ \nabla\cdot\mathbf{B}=0 \\ &\text{(iv)}\ \nabla\times\mathbf{H}=\frac{\partial\mathbf{D}}{\partial t} \end{aligned} \tag{9.66}$$

Linear constitutive relations: $\mathbf{D}=\epsilon\mathbf{E}$, $\mathbf{H}=\mathbf{B}/\mu$ Eq. 9.67. For a homogeneous medium:

$$\begin{aligned} &\text{(i)}\ \nabla\cdot\mathbf{E}=0 \\ &\text{(iii)}\ \nabla\times\mathbf{E}=-\frac{\partial\mathbf{B}}{\partial t} \\ &\text{(ii)}\ \nabla\cdot\mathbf{B}=0 \\ &\text{(iv)}\ \nabla\times\mathbf{B}=\mu\epsilon\frac{\partial\mathbf{E}}{\partial t} \end{aligned} \tag{9.68}$$

2. Speed and index

PLUS TWO RECALL: REFRACTIVE INDEX

Recall. From Class 12: $n=c/v$.

Re-derive.

1. Replacing $\mu_0\epsilon_0$ by $\mu\epsilon$ in the vacuum wave equation gives $v^2=1/\mu\epsilon$.
2. Writing $v=c/n$, $n=c/v=\sqrt{\mu\epsilon/\mu_0\epsilon_0}$.

Saved. $v=c/n$ Eq. 9.69.

$$v=\frac{1}{\sqrt{\epsilon\mu}}=\frac{c}{n} \tag{9.69}$$

$$n\equiv\sqrt{\frac{\epsilon\mu}{\epsilon_0\mu_0}}$$

$$n\approx\sqrt{\epsilon_r} \tag{9.71}$$

Index of refraction Eq. 9.70 and the dielectric constant ϵ_r (Eq. 4.34). Since $\epsilon_r>1$, light is slower in matter. Footnote 13: the induced dipole fields combine with the original field into a single wave with a new speed (the Ewald–Oseen theorem); transparency follows. Footnote 14: ϵ_r may depend on frequency, so static values give the wrong n (water: 8.9 static, 1.33 for visible light).

3. Energy and flux

$$u=\frac{1}{2}\left(\epsilon E^2+\frac{1}{\mu}B^2\right) \tag{9.72}$$

$$\mathbf{S}=\frac{1}{\mu}(\mathbf{E}\times\mathbf{B}) \tag{9.73}$$

Plane wave: $\omega=kv$ Eq. 9.11, $B_0=E_0/v$ (9.48 with $c\to v$), and

RESULT 9.11: INTENSITY IN A MEDIUM

$$I=\frac{1}{2}\epsilon v E_0^2 \tag{9.74}$$

4. Interface conditions

$$\begin{aligned} &\text{(i)}\ \epsilon_1E_1^\perp=\epsilon_2E_2^\perp \\ &\text{(iii)}\ \mathbf{E}_1^\parallel=\mathbf{E}_2^\parallel \\ &\text{(ii)}\ B_1^\perp=B_2^\perp \\ &\text{(iv)}\ \frac{1}{\mu_1}\mathbf{B}_1^\parallel=\frac{1}{\mu_2}\mathbf{B}_2^\parallel \end{aligned} \tag{9.75}$$

5. Normal incidence

Interface in the xy plane. Incident, reflected and transmitted waves (Fig. 9.13):

$$\tilde{\mathbf{E}}_I = \tilde{E}_{0I} e^{i(k_1 z - \omega t)} \hat{x}$$
$$\tilde{\mathbf{B}}_I = \frac{1}{v_1} \tilde{E}_{0I} e^{i(k_1 z - \omega t)} \hat{y} \tag{9.76}$$

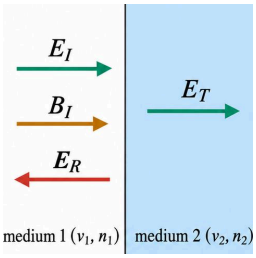


Fig. 9.13 (book): normal incidence on the interface between medium 1 and medium 2.

$$\tilde{\mathbf{E}}_R = \tilde{E}_{0R} e^{i(-k_1 z - \omega t)} \hat{x}$$
$$\tilde{\mathbf{B}}_R = -\frac{1}{v_1} \tilde{E}_{0R} e^{i(-k_1 z - \omega t)} \hat{y} \tag{9.77}$$

$$\tilde{\mathbf{E}}_T = \tilde{E}_{0T} e^{i(k_2 z - \omega t)} \hat{x}$$
$$\tilde{\mathbf{B}}_T = \frac{1}{v_2} \tilde{E}_{0I} e^{i(k_2 z - \omega t)} \hat{y} \tag{9.78}$$

Minus sign in $\tilde{\mathbf{B}}_R$: required by Eq. 9.50. Conditions (iii) and (iv) at $z = 0$:

$$\tilde{E}_{0I} + \tilde{E}_{0R} = \tilde{E}_{0T} \tag{9.79}$$

$$\frac{1}{\mu_1}\left(\frac{1}{v_1}\tilde{E}_{0I} - \frac{1}{v_1}\tilde{E}_{0R}\right) = \frac{1}{\mu_2}\frac{1}{v_2}\tilde{E}_{0T} \tag{9.80}$$

Writing $\tilde{E}_{0I} - \tilde{E}_{0R} = \beta \tilde{E}_{0T}$ Eq. 9.81, $\beta \equiv \mu_1 v_1 / \mu_2 v_2 = \mu_1 n_2 / \mu_2 n_1$ Eq. 9.82. Solving:

RESULT 9.12: NORMAL-INCIDENCE AMPLITUDES

$$\tilde{E}_{0R} = \left(\frac{1-\beta}{1+\beta}\right)\tilde{E}_{0I}$$
$$\tilde{E}_{0T} = \left(\frac{2}{1+\beta}\right)\tilde{E}_{0I} \tag{9.83}$$

$$\tilde{E}_{0R} = \left(\frac{v_2 - v_1}{v_2 + v_1}\right)\tilde{E}_{0I}$$
$$\tilde{E}_{0T} = \left(\frac{2v_2}{v_2 + v_1}\right)\tilde{E}_{0I} \tag{9.84}$$

Real amplitudes Eq. 9.85, and in terms of indices Eq. 9.86:

$$E_{0R} = \left|\frac{v_2 - v_1}{v_2 + v_1}\right| E_{0I}$$
$$E_{0T} = \left(\frac{2v_2}{v_2 + v_1}\right) E_{0I} \tag{9.85}$$

$$E_{0R} = \left|\frac{n_1 - n_2}{n_1 + n_2}\right| E_{0I}$$
$$E_{0T} = \left(\frac{2n_1}{n_1 + n_2}\right) E_{0I} \tag{9.86}$$

6. Coefficients

$$R = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2 \tag{9.87}$$

$$T = \frac{4 n_1 n_2}{(n_1 + n_2)^2} \tag{9.88}$$

$$R + T = 1 \tag{9.89}$$

Check: air to glass gives $R = 0.04$, $T = 0.96$. Problems 9.14 and 9.15 are outside these notes.

9.3.3 Reflection and transmission at oblique incidence

Source: book-sources/m3-sec4-oblique.html, Griffiths pp. 411–418. Equations 9.90–9.118 in full; footnotes 17–19. ***Background*** *(beyond the syllabus list). Problems 9.16–9.19 listed, not worked.*

Prerequisites

- Normal incidence and the interface conditions (§9.3.2).
- Snell's law (pre-physics P5); wave vectors and $\mathbf{k}\cdot\mathbf{r}$ (pre-math M10).

1. Waves

$$\tilde{\mathbf{E}}_I = \tilde{\mathbf{E}}_{0I} e^{i(\mathbf{k}_I\cdot\mathbf{r}-\omega t)}$$
$$\tilde{\mathbf{B}}_I = \frac{1}{v_1}(\hat{k}_I \times \tilde{\mathbf{E}}_I) \tag{9.90}$$

$$\tilde{\mathbf{E}}_R = \tilde{\mathbf{E}}_{0R} e^{i(\mathbf{k}_R\cdot\mathbf{r}-\omega t)}$$
$$\tilde{\mathbf{B}}_R = \frac{1}{v_1}(\hat{k}_R \times \tilde{\mathbf{E}}_R) \tag{9.91}$$

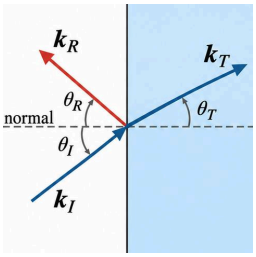


Fig. 9.14 (book): oblique incidence; wave vectors k_I, k_R, k_T and angles θ_I, θ_R, θ_T to the normal.

$$\tilde{\mathbf{E}}_T = \tilde{\mathbf{E}}_{0T} e^{i(\mathbf{k}_T\cdot\mathbf{r}-\omega t)}$$
$$\tilde{\mathbf{B}}_T = \frac{1}{v_2}(\hat{k}_T \times \tilde{\mathbf{E}}_T) \tag{9.92}$$

Common frequency ω (footnote 17: linear media only). Wave numbers:

$$k_I v_1 = k_R v_1 = k_T v_2 = \omega$$
$$k_I = k_R = \frac{v_2}{v_1}k_T = \frac{n_1}{n_2}k_T \tag{9.93}$$

2. Matching exponentials

The matching condition Eq. 9.75 must hold for all x, y and t. So the exponential factors must agree on the plane $z = 0$:

$$\mathbf{k}_I\cdot\mathbf{r} = \mathbf{k}_R\cdot\mathbf{r} = \mathbf{k}_T\cdot\mathbf{r} \quad \text{when } z = 0 \tag{9.95}$$

$$x(k_I)_x + y(k_I)_y = x(k_R)_x + y(k_R)_y = x(k_T)_x + y(k_T)_y \tag{9.96}$$

Separating the x and y coefficients:

$$(k_I)_y = (k_R)_y = (k_T)_y \tag{9.97}$$
$$(k_I)_x = (k_R)_x = (k_T)_x \tag{9.98}$$

3. Laws of geometrical optics

RESULT 9.13: THE THREE LAWS

First law. The three wave vectors and the normal lie in one plane, the plane of incidence.

Second law. From Eq. 9.98, $k_I \sin\theta_I = k_R \sin\theta_R = k_T \sin\theta_T$ Eq. 9.99, so

$$\theta_I = \theta_R \tag{9.100}$$

Third law (Snell).

$$\frac{\sin\theta_T}{\sin\theta_I} = \frac{n_1}{n_2} \tag{9.101}$$

Only the general form of the boundary condition Eq. 9.94 was used. So these laws hold for any wave that crosses an interface, including water and sound waves.

4. Amplitude conditions

$$\epsilon_1(\tilde{\mathbf{E}}_{0I} + \tilde{\mathbf{E}}_{0R})_z = \epsilon_2(\tilde{\mathbf{E}}_{0T})_z$$
$$(\tilde{\mathbf{B}}_{0I} + \tilde{\mathbf{B}}_{0R})_z = (\tilde{\mathbf{B}}_{0T})_z \tag{9.102}$$

together with the x- and y-components of $(\tilde{\mathbf{E}}_{0I} + \tilde{\mathbf{E}}_{0R})$ and of $\frac{1}{\mu}(\tilde{\mathbf{B}}_{0I} + \tilde{\mathbf{B}}_{0R})$ (9.102, pairs of equations).

5. Polarization parallel to the plane of incidence

$$\epsilon_1(-\tilde{E}_{0I}\sin\theta_I + \tilde{E}_{0R}\sin\theta_R) = \epsilon_2(-\tilde{E}_{0T}\sin\theta_T) \tag{9.103}$$
$$\tilde{E}_{0I}\cos\theta_I + \tilde{E}_{0R}\cos\theta_R = \tilde{E}_{0T}\cos\theta_T \tag{9.104}$$
$$\frac{1}{\mu_1 v_1}(\tilde{E}_{0I} - \tilde{E}_{0R}) = \frac{1}{\mu_2 v_2}\tilde{E}_{0T} \tag{9.105}$$

Using the laws above, Eq. 9.103 and Eq. 9.105 give

$$\tilde{E}_{0I} - \tilde{E}_{0R} = \beta\tilde{E}_{0T} \tag{9.106}$$
$$\beta \equiv \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{\mu_1 n_2}{\mu_2 n_1} \tag{9.107}$$

and Eq. 9.104 gives

$$\tilde{E}_{0I} + \tilde{E}_{0R} = \alpha\tilde{E}_{0T} \tag{9.108}$$
$$\alpha \equiv \frac{\cos\theta_T}{\cos\theta_I} \tag{9.109}$$

RESULT 9.14: FRESNEL EQUATIONS, PARALLEL POLARIZATION

$$\begin{aligned}\tilde{E}_{0R} &= \left(\frac{\alpha-\beta}{\alpha+\beta}\right)\tilde{E}_{0I} \\ \tilde{E}_{0T} &= \left(\frac{2}{\alpha+\beta}\right)\tilde{E}_{0I}\end{aligned} \tag{9.110}$$

$$\begin{aligned}\alpha &= \frac{\sqrt{1-\sin^2\theta_T}}{\cos\theta_I} \\ &= \frac{\sqrt{1-[(n_1/n_2)\sin\theta_I]^2}}{\cos\theta_I}\end{aligned} \tag{9.111}$$

Normal incidence gives $\alpha = 1$ and recovers Eq. 9.83. Grazing incidence gives total reflection.

6. Brewster's angle

The reflected amplitude vanishes when $\alpha = \beta$, which gives

$$\sin^2\theta_B = \frac{1-\beta^2}{(n_1/n_2)^2-\beta^2} \tag{9.112}$$

For $\mu_1 \approx \mu_2$: $\tan\theta_B \approx n_2/n_1$ Eq. 9.113. The reflected beam is then polarized perpendicular to the plane of incidence (footnote 19).

7. Intensities and coefficients

$$I_I = \frac{1}{2}\epsilon_1 v_1 E_{0I}^2 \cos\theta_I \tag{9.114}$$

$$\begin{aligned}I_R &= \frac{1}{2}\epsilon_1 v_1 E_{0R}^2 \cos\theta_R \\ I_T &= \frac{1}{2}\epsilon_2 v_2 E_{0T}^2 \cos\theta_T\end{aligned} \tag{9.115}$$

RESULT 9.15: COEFFICIENTS (PARALLEL POLARIZATION)

$$R = \left(\frac{\alpha-\beta}{\alpha+\beta}\right)^2 \tag{9.116}$$

$$T = \alpha\beta\left(\frac{2}{\alpha+\beta}\right)^2 \tag{9.117}$$

$$I_I = I_T + I_R \tag{9.118}$$

Hence $R+T=1$. R vanishes and T reaches 1 at Brewster's angle.

Problems 9.16–9.19 are outside these notes.

Section 5, background beyond the syllabus list: absorption, dispersion and guided waves (Griffiths 9.4, 9.5)

9.4 Absorption and dispersion (background)

*Source: book-sources/m3-sec4-oblique.html (p. 418), m3-bg-absorption.html, Griffiths pp. 418–430. **Background**, beyond the syllabus list. Equations 9.119–9.176 in full. Problems 9.20–9.27 listed, not worked.*

Prerequisites

- Complex wave numbers and the real part (§9.1.2).
- Ohm's law (Module 1 and Plus Two current electricity).

1. Conductors (§9.4.1)

$$\mathbf{J}_f = \sigma \mathbf{E} \tag{9.119}$$

$$\begin{aligned} \text{(i)}\ \nabla\cdot\mathbf{E} &= \frac{1}{\epsilon}\rho_f \\ \text{(iii)}\ \nabla\times\mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \text{(ii)}\ \nabla\cdot\mathbf{B} &= 0 \\ \text{(iv)}\ \nabla\times\mathbf{B} &= \mu\sigma\mathbf{E} + \mu\epsilon\frac{\partial \mathbf{E}}{\partial t} \end{aligned} \tag{9.120}$$

Continuity of free charge gives $\nabla\cdot\mathbf{J}_f = -\partial\rho_f/\partial t$ Eq. 9.121, and with Ohm's and Gauss's laws

$$\rho_f(t) = e^{-(\sigma/\epsilon)t}\rho_f(0) \tag{9.122}$$

so free charge relaxes to zero in $\tau = \epsilon/\sigma$ (footnote 20). Then $\rho_f = 0$ and

$$\begin{aligned} \text{(i)}\ \nabla\cdot\mathbf{E} &= 0 \\ \text{(iii)}\ \nabla\times\mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \text{(ii)}\ \nabla\cdot\mathbf{B} &= 0 \\ \text{(iv)}\ \nabla\times\mathbf{B} &= \mu\epsilon\frac{\partial \mathbf{E}}{\partial t} + \mu\sigma\mathbf{E} \end{aligned} \tag{9.123}$$

RESULT 9.16: MODIFIED WAVE EQUATIONS

$$\begin{aligned} \nabla^2\mathbf{E} &= \mu\epsilon\frac{\partial^2\mathbf{E}}{\partial t^2} + \mu\sigma\frac{\partial \mathbf{E}}{\partial t} \\ \nabla^2\mathbf{B} &= \mu\epsilon\frac{\partial^2\mathbf{B}}{\partial t^2} + \mu\sigma\frac{\partial \mathbf{B}}{\partial t} \end{aligned} \tag{9.124}$$

Plane-wave solutions $\tilde{\mathbf{E}} = \tilde{\mathbf{E}}_0 e^{i(\tilde{k}z-\omega t)}$ Eq. 9.125 require $\tilde{k}^2 = \mu\epsilon\omega^2 + i\mu\sigma\omega$ Eq. 9.126. With $\tilde{k} = k + i\kappa$ Eq. 9.127:

$$\begin{aligned} k &\equiv \omega\sqrt{\frac{\epsilon\mu}{2}}\left[\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}+1\right]^{1/2} \\ \kappa &\equiv \omega\sqrt{\frac{\epsilon\mu}{2}}\left[\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}-1\right]^{1/2} \end{aligned} \tag{9.128}$$

Attenuated fields Eq. 9.129, skin depth $d = 1/\kappa$ Eq. 9.130, and

$$\begin{aligned} \lambda &= \frac{2\pi}{k} \\ v &= \frac{\omega}{k} \\ n &= \frac{ck}{\omega} \end{aligned} \tag{9.131}$$

With $\mathbf{E}$ along x Eq. 9.132, $\tilde{\mathbf{B}} = (\tilde{k}/\omega)\tilde{E}_0 e^{-\kappa z}e^{i(kz-\omega t)}\hat{y}$ Eq. 9.133. Polar form $\tilde{k} = Ke^{i\phi}$ Eq. 9.134, with

$$K \equiv |\tilde{k}| = \omega\sqrt{\epsilon\mu\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}} \tag{9.135}$$

$$\phi \equiv \tan^{-1}(\kappa/k) \tag{9.136}$$

Hence the complex amplitudes are related by $B_0 e^{i\delta_B} = (Ke^{i\phi}/\omega)E_0 e^{i\delta_E}$ Eq. 9.137, and the phase difference and amplitude ratio are

$$\delta_B - \delta_E = \phi \tag{9.138}$$

$$\begin{aligned} \frac{B_0}{E_0} &= \frac{K}{\omega} \\ &= \sqrt{\epsilon\mu\sqrt{1+\left(\frac{\sigma}{\epsilon\omega}\right)^2}} \end{aligned} \tag{9.139}$$

RESULT 9.17: REAL FIELDS IN A CONDUCTOR

$$\begin{aligned} \mathbf{E}(z,t) &= E_0 e^{-\kappa z}\cos(kz-\omega t+\delta_E)\hat{x} \\ \mathbf{B}(z,t) &= B_0 e^{-\kappa z}\cos(kz-\omega t+\delta_E+\phi)\hat{y} \end{aligned} \tag{9.14}$$

2. Reflection at a conducting surface (§9.4.2)

$$\begin{aligned} \text{(i)}\ \epsilon_1 E_1^{\perp} - \epsilon_2 E_2^{\perp} &= \sigma_f \\ \text{(iii)}\ \mathbf{E}_1^{\parallel} - \mathbf{E}_2^{\parallel} &= 0 \\ \text{(ii)}\ B_1^{\perp} - B_2^{\perp} &= 0 \\ \text{(iv)}\ \frac{1}{\mu_1}\mathbf{B}_1^{\parallel} - \frac{1}{\mu_2}\mathbf{B}_2^{\parallel} &= \mathbf{K}_f\times\hat{\mathbf{n}} \end{aligned} \tag{9.141}$$

$$\tilde{\mathbf{E}}_I = \tilde{E}_{0I} e^{i(k_1 z - \omega t)} \hat{x}$$
$$\tilde{\mathbf{B}}_I = \frac{1}{v_1} \tilde{E}_{0I} e^{i(k_1 z - \omega t)} \hat{y} \tag{9.142}$$

$$\tilde{\mathbf{E}}_R = \tilde{E}_{0R} e^{i(-k_1 z - \omega t)} \hat{x}$$
$$\tilde{\mathbf{B}}_R = -\frac{1}{v_1} \tilde{E}_{0R} e^{i(-k_1 z - \omega t)} \hat{y} \tag{9.143}$$

$$\tilde{\mathbf{E}}_T = \tilde{E}_{0T} e^{i(\tilde{k}_2 z - \omega t)} \hat{x}$$
$$\tilde{\mathbf{B}}_T = \frac{\tilde{k}_2}{\omega} \tilde{E}_{0T} e^{i(\tilde{k}_2 z - \omega t)} \hat{y} \tag{9.144}$$

$$\tilde{E}_{0I} + \tilde{E}_{0R} = \tilde{E}_{0T} \tag{9.145}$$

$$\frac{1}{\mu_1 v_1}(\tilde{E}_{0I} - \tilde{E}_{0R}) - \frac{\tilde{k}_2}{\mu_2 \omega} \tilde{E}_{0T} = 0 \tag{9.146}$$

$$\tilde{E}_{0I} - \tilde{E}_{0R} = \tilde{\beta} \tilde{E}_{0T}$$
$$\tilde{\beta} \equiv \frac{\mu_1 v_1}{\mu_2 \omega} \tilde{k}_2 \tag{9.147}$$

$$\tilde{E}_{0R} = \left(\frac{1 - \tilde{\beta}}{1 + \tilde{\beta}} \right) \tilde{E}_{0I}$$
$$\tilde{E}_{0T} = \left(\frac{2}{1 + \tilde{\beta}} \right) \tilde{E}_{0I} \tag{9.149}$$

Here $\tilde{\beta}$ is defined by Eq. 9.148. For a perfect conductor, $\tilde{E}_{0R} = -\tilde{E}_{0I}$ and $\tilde{E}_{0T} = 0$ Eq. 9.150. Problem 9.23 is outside these notes.

3. Dispersion (§9.4.3)

$$v = \frac{\omega}{k} \tag{9.151}$$

$$v_g = \frac{d\omega}{dk} \tag{9.152}$$

Model: spring force $-m\omega_0^2 x$ Eq. 9.153, damping $-m\gamma\, dx/dt$ Eq. 9.154, driving $qE_0 \cos \omega t$ Eq. 9.155, equation of motion Eq. 9.156. Complex steady state Eq. 9.157 with $\tilde{x} = \tilde{x}_0 e^{-i\omega t}$ Eq. 9.158 gives

$$\tilde{x}_0 = \frac{q/m}{\omega_0^2 - \omega^2 - i\gamma\omega} E_0 \tag{9.159}$$

$$\tilde{p}(t) = q\tilde{x}(t)$$
$$= \frac{q^2/m}{\omega_0^2 - \omega^2 - i\gamma\omega} E_0 e^{-i\omega t} \tag{9.160}$$

Summing over electrons of each type with N molecules per unit volume:

$$\tilde{\mathbf{P}} = \frac{Nq^2}{m} \left(\sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j \omega} \right) \tilde{\mathbf{E}} \tag{9.161}$$

$$\tilde{\mathbf{P}} = \epsilon_0 \tilde{\chi}_e \tilde{\mathbf{E}} \tag{9.162}$$

$$\tilde{\epsilon}_r = \frac{\tilde{\epsilon}}{\epsilon_0}$$
$$= 1 + \frac{Nq^2}{m\epsilon_0} \sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j \omega} \tag{9.163}$$

$$\nabla^2 \tilde{\mathbf{E}} = \tilde{\epsilon} \mu_0 \frac{\partial^2 \tilde{\mathbf{E}}}{\partial t^2} \tag{9.164}$$

$$\tilde{\mathbf{E}}(z,t) = \tilde{\mathbf{E}}_0 e^{i(\tilde{k} z - \omega t)} \tag{9.165}$$

$$\tilde{k} \equiv \sqrt{\tilde{\epsilon} \mu_0}\, \omega \tag{9.166}$$

$$\tilde{k} = k + i\kappa \tag{9.167}$$

$$\tilde{\mathbf{E}}(z,t) = \tilde{\mathbf{E}}_0 e^{-\kappa z} e^{i(kz - \omega t)} \tag{9.168}$$

$$\alpha \equiv 2\kappa \tag{9.169}$$

$$n = \frac{ck}{\omega} \tag{9.170}$$

For gases, $\sqrt{1+\varepsilon} \approx 1 + \frac{1}{2}\varepsilon$ gives

$$\tilde{k} = \frac{\omega}{c}\sqrt{\tilde{\epsilon}_r} \approx \frac{\omega}{c}\left[1 + \frac{Nq^2}{2m\epsilon_0} \sum_j \frac{f_j}{\omega_j^2 - \omega^2 - i\gamma_j \omega} \right] \tag{9.171}$$

$$n \approx 1 + \frac{Nq^2}{2m\epsilon_0} \sum_j \frac{f_j(\omega_j^2 - \omega^2)}{(\omega_j^2 - \omega^2)^2 + \gamma_j^2 \omega^2} \tag{9.172}$$

$$\alpha \approx \frac{Nq^2 \omega^2}{m\epsilon_0 c} \sum_j \frac{f_j \gamma_j}{(\omega_j^2 - \omega^2)^2 + \gamma_j^2 \omega^2} \tag{9.173}$$

Normal dispersion (index rising with ω) holds away from resonance; anomalous dispersion occurs near each resonance (Fig. 9.22). With damping neglected and $\omega < \omega_j$:

$$n = 1 + \frac{Nq^2}{2m\epsilon_0} \sum_j \frac{f_j}{\omega_j^2 - \omega^2} \tag{9.174}$$

$$n = 1 + \left(\frac{Nq^2}{2m\epsilon_0} \sum_j \frac{f_j}{\omega_j^2} \right) + \omega^2 \left(\frac{Nq^2}{2m\epsilon_0} \sum_j \frac{f_j}{\omega_j^4} \right) \tag{9.175}$$

RESULT 9.18: CAUCHY'S FORMULA

$$n = 1 + A\left(1 + \frac{B}{\lambda^2}\right) \tag{9.176}$$

A is the coefficient of refraction and B the coefficient of dispersion. Problems 9.24–9.27 are outside these notes.

9.5 Guided waves (background)

*Source: book-sources/m3-bg-guided.html (and p. 430 in m3-bg-absorption.html), Griffiths pp. 430–441. **Background**, beyond the syllabus list. Equations 9.177–9.208 in full. Problems 9.28–9.44 listed, not worked.*

Prerequisites

- Maxwell's equations in vacuum (Eq. 9.40).
- Vector identities for the divergence and curl (pre-math M13).
- Separation of variables in a partial differential equation.

1. Hollow wave guides (§9.5.1)

Perfect conductor walls. At the inner surface:

$$\text{(i) } \mathbf{E}^{\parallel} = \mathbf{0}, \qquad \text{(ii) } B^{\perp} = 0 \tag{9.177}$$

Monochromatic fields travelling down the guide, with real k (footnote 27 on surface charges):

$$\begin{aligned} \text{(i) } \tilde{\mathbf{E}}(x,y,z,t) &= \tilde{\mathbf{E}}_0(x,y)e^{i(kz-\omega t)} \\ \text{(ii) } \tilde{\mathbf{B}}(x,y,z,t) &= \tilde{\mathbf{B}}_0(x,y)e^{i(kz-\omega t)} \end{aligned} \tag{9.178}$$

Maxwell's equations in the interior:

$$\begin{aligned} &\text{(i) } \nabla\cdot\mathbf{E} = 0 \\ &\text{(iii) } \nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t} \\ &\text{(ii) } \nabla\cdot\mathbf{B} = 0 \\ &\text{(iv) } \nabla\times\mathbf{B} = \frac{1}{c^2}\frac{\partial\mathbf{E}}{\partial t} \end{aligned} \tag{9.179}$$

Confined waves are not transverse in general. The longitudinal components are needed to satisfy (9.177):

$$\begin{aligned} \tilde{\mathbf{E}}_0 &= E_x\hat{x} + E_y\hat{y} + E_z\hat{z} \\ \tilde{\mathbf{B}}_0 &= B_x\hat{x} + B_y\hat{y} + B_z\hat{z} \end{aligned} \tag{9.180}$$

Substituting into (iii) and (iv) gives six relations (9.181):

$$\begin{aligned} &\text{(i) } \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = i\omega B_z \\ &\text{(iv) } \frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} = -\frac{i\omega}{c^2}E_z \\ &\text{(ii) } \frac{\partial E_z}{\partial y} - ikE_y = i\omega B_x \\ &\text{(v) } \frac{\partial B_z}{\partial y} - ikB_y = -\frac{i\omega}{c^2}E_x \\ &\text{(iii) } ikE_x - \frac{\partial E_z}{\partial x} = i\omega B_y \\ &\text{(vi) } ikB_x - \frac{\partial B_z}{\partial x} = -\frac{i\omega}{c^2}E_y \end{aligned} \tag{9.181}$$

Four of these solve for the transverse components (9.182):

$$\begin{aligned} &\text{(i) } E_x = \frac{i}{(\omega/c)^2 - k^2}\left(k\frac{\partial E_z}{\partial x} + \omega\frac{\partial B_z}{\partial y}\right) \\ &\text{(ii) } E_y = \frac{i}{(\omega/c)^2 - k^2}\left(k\frac{\partial E_z}{\partial y} - \omega\frac{\partial B_z}{\partial x}\right) \\ &\text{(iii) } B_x = \frac{i}{(\omega/c)^2 - k^2}\left(k\frac{\partial B_z}{\partial x} - \frac{\omega}{c^2}\frac{\partial E_z}{\partial y}\right) \\ &\text{(iv) } B_y = \frac{i}{(\omega/c)^2 - k^2}\left(k\frac{\partial B_z}{\partial y} + \frac{\omega}{c^2}\frac{\partial E_z}{\partial x}\right) \end{aligned} \tag{9.182}$$

So only E_z and B_z need solving. They satisfy (9.183):

$$\begin{aligned} &\text{(i) } \left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + (\omega/c)^2 - k^2\right]E_z = 0 \\ &\text{(ii) } \left[\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + (\omega/c)^2 - k^2\right]B_z = 0 \end{aligned} \tag{9.183}$$

$E_z = 0$: TE. $B_z = 0$: TM. Both zero: TEM, which a hollow guide cannot carry (footnote 29). Eq. 9.181 is indeterminate for TEM waves, where $k = \omega/c$.

2. TE modes of a rectangular guide (§9.5.2)

Separation of variables, $B_z = X(x)Y(y)$, gives (9.184) and (9.185):

$$\begin{aligned} &\text{(i) } \frac{1}{X}\frac{d^2X}{dx^2} = -k_x^2 \\ &\text{(ii) } \frac{1}{Y}\frac{d^2Y}{dy^2} = -k_y^2 \end{aligned} \tag{9.184}$$

$$-k_x^2 - k_y^2 + (\omega/c)^2 - k^2 = 0 \tag{9.185}$$

RESULT 9.19: ALLOWED TRANSVERSE WAVE NUMBERS

$$k_x = \frac{m\pi}{a} \tag{9.186}$$

$$k_y = \frac{n\pi}{b} \tag{9.187}$$

$$B_z = B_0\cos(m\pi x/a)\cos(n\pi y/b) \tag{9.188}$$

This is the TE$_{mn}$ mode (with $a \geq b$, and at least one of m, n nonzero). From (9.185):

$$k = \sqrt{(\omega/c)^2 - \pi^2\left[(m/a)^2 + (n/b)^2\right]} \tag{9.189}$$

Cutoff: if $\omega < \omega_{mn}$ the fields are attenuated (9.190):

$$\omega_{mn} \equiv c\pi\sqrt{(m/a)^2 + (n/b)^2} \tag{9.190}$$

The lowest cutoff is

$$\omega_{10} = c\pi/a \tag{9.191}$$

The wave number in terms of cutoff is (9.192):

$$k = \frac{1}{c}\sqrt{\omega^2 - \omega_{mn}^2} \tag{9.192}$$

$$v = \frac{\omega}{k} = \frac{c}{\sqrt{1 - (\omega_{mn}/\omega)^2}} \tag{9.193}$$

$$v_g = \frac{1}{dk/d\omega} = c\sqrt{1 - (\omega_{mn}/\omega)^2} < c \tag{9.194}$$

The phase velocity v exceeds c, but energy moves at $v_g < c$ (Prob. 9.31). Figure 9.25 (book) shows the same result as a plane wave bouncing between the walls at angle θ, with $\mathbf{k}' = (\pi m/a)\hat{x} + (\pi n/b)\hat{y} + k\hat{z}$ and $\cos\theta = k/|\mathbf{k}'|$.

3. The coaxial line (§9.5.3)

The central wire and the sheath allow $E_z = B_z = 0$. Then (9.181) gives

$$k = \frac{\omega}{c} \tag{9.195}$$

$$cB_y = E_x, \qquad cB_x = -E_y \tag{9.196}$$

and the transverse fields obey the two-dimensional electrostatic and magnetostatic equations (9.197):

$$\begin{aligned} \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} &= 0 \\ \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} &= 0 \\ \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} &= 0 \\ \frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} &= 0 \end{aligned} \tag{9.197}$$

The solution with cylindrical symmetry (9.198) is borrowed from the line charge and the straight current:

$$\begin{aligned} \mathbf{E}_0(s,\phi) &= \frac{A}{s}\hat{s} \\ \mathbf{B}_0(s,\phi) &= \frac{A}{cs}\hat{\phi} \end{aligned} \tag{9.198}$$

$$\begin{aligned} \mathbf{E}(s,\phi,z,t) &= \frac{A\cos(kz-\omega t)}{s}\hat{s} \\ \mathbf{B}(s,\phi,z,t) &= \frac{A\cos(kz-\omega t)}{cs}\hat{\phi} \end{aligned} \tag{9.199}$$

4. Chapter-end problems used in the notes

The equations in the “More Problems on Chapter 9” section are given in the book source. Those used in these notes are: the Fourier inversion theorem (Prob. 9.34, Eq. 9.200); the slab transmission coefficient (Prob. 9.38, Eq. 9.201); the critical angle (Prob. 9.41, Eq. 9.202); the evanescent wave (Prob. 9.41, Eqs. 9.203–9.205); the resonant cavity (Prob. 9.42, Eq. 9.206); and optical tweezers (Prob. 9.44, Eqs. 9.207–9.208). Problems are not worked here.

$$\begin{aligned} \tilde{\phi}(z) &= \int_{-\infty}^{\infty} \tilde{\Phi}(k) e^{ikz}\, dk \quad \Leftrightarrow \quad \tilde{\Phi}(k) \\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{\phi}(z) e^{-ikz}\, dz \end{aligned} \tag{9.200}$$

$$\begin{aligned} T^{-1} = \frac{1}{4n_1 n_3}\Big[&(n_1+n_3)^2 \\ &+ \frac{(n_1^2-n_2^2)(n_2^2-n_3^2)}{n_2^2}\sin^2\Big(\frac{n_2\omega d}{c}\Big)\Big] \end{aligned} \tag{9.201}$$

$$\theta_c \equiv \sin^{-1}(n_2/n_1) \tag{9.202}$$

$$\tilde{\mathbf{E}}_T(\mathbf{r},t) = \tilde{\mathbf{E}}_{0T} e^{-\kappa z} e^{i(kx-\omega t)} \tag{9.203}$$

$$\begin{aligned} \kappa &\equiv \frac{\omega}{c}\sqrt{(n_1\sin\theta_I)^2 - n_2^2} \\ &\text{and} \\ k &\equiv \frac{\omega n_1}{c}\sin\theta_I \end{aligned} \tag{9.204}$$

$$\begin{aligned} \mathbf{E}(\mathbf{r},t) &= E_0 e^{-\kappa z}\cos(kx-\omega t)\hat{y}, \\ \mathbf{B}(\mathbf{r},t) &= \frac{E_0}{\omega} e^{-\kappa z}\left[\kappa\sin(kx-\omega t)\hat{x} + k\cos(kx-\omega t)\hat{z}\right] \end{aligned} \tag{9.20}$$

$$\omega_{lmn} = c\pi\sqrt{(l/d)^2 + (m/a)^2 + (n/b)^2} \tag{9.206}$$

$$\mathbf{F} = \alpha\left[\frac{1}{2}\nabla E^2 + \frac{\partial}{\partial t}(\mathbf{E}\times\mathbf{B})\right] \tag{9.207}$$

$$\mathbf{F} = \frac{\alpha}{2}\nabla E^2 \tag{9.208}$$